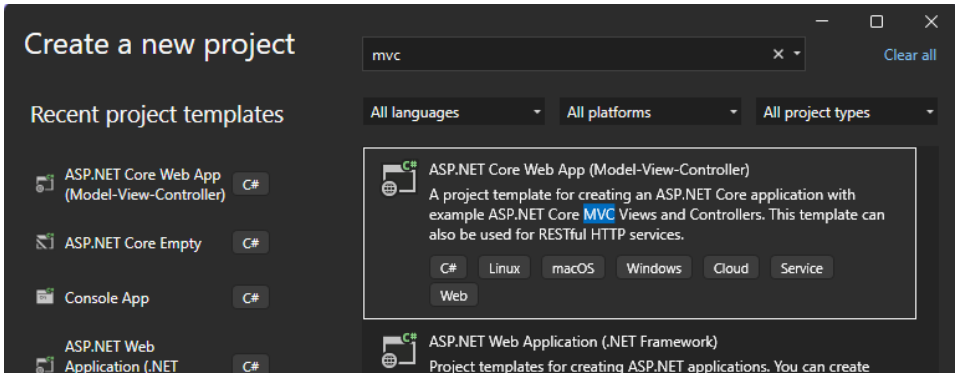
a. Create an MVC application to perform basic **Arithmetic Operations**.

Step 1: Open Visual Studio 2026->Click Create a new project.

Step 2: Select Project Template



Note: If this template is not visible, install the **ASP.NET and Web Development** workload from the Visual Studio Installer.

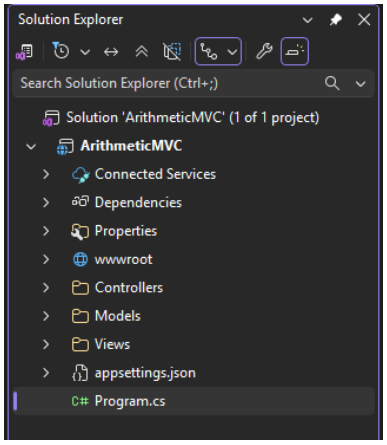
Step 3: Configure the Project. Project Name: ArithmeticMVC

Step 4: Select MVC Template

A dialog box named **Create a new ASP.NET Web Application** will appear.

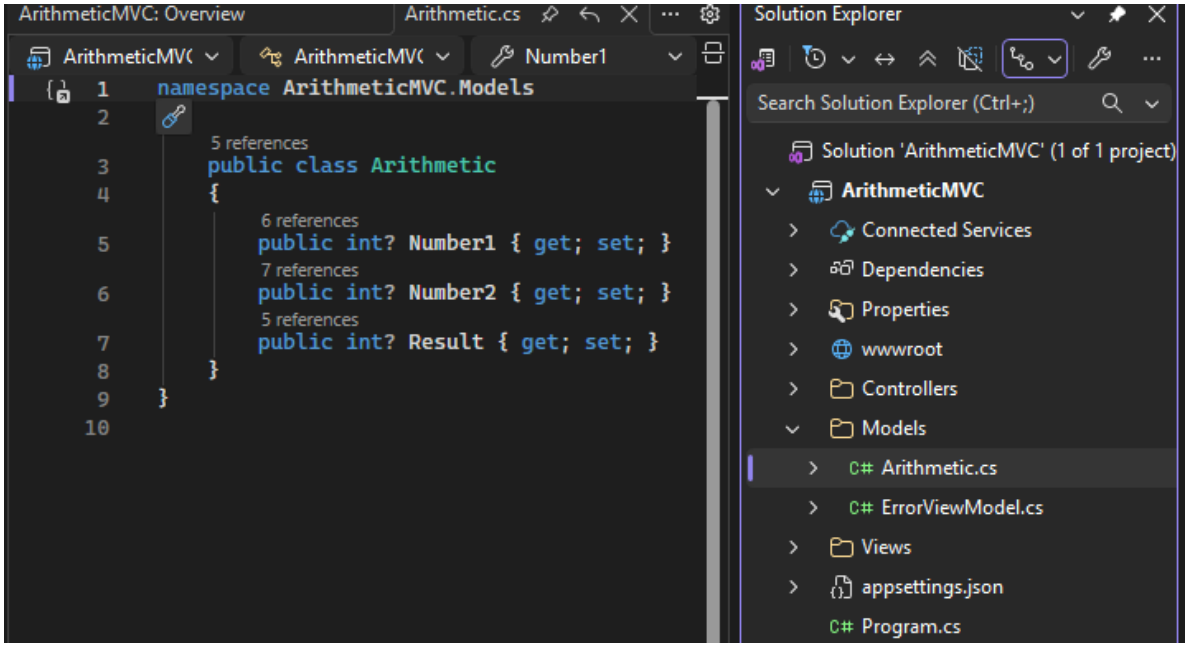
- Select **MVC** → Authentication: **No Authentication** → Leave **Configure for HTTPS** checked (optional). → Click **Create** → Visual Studio creates the following folders

ArithmeticMVC



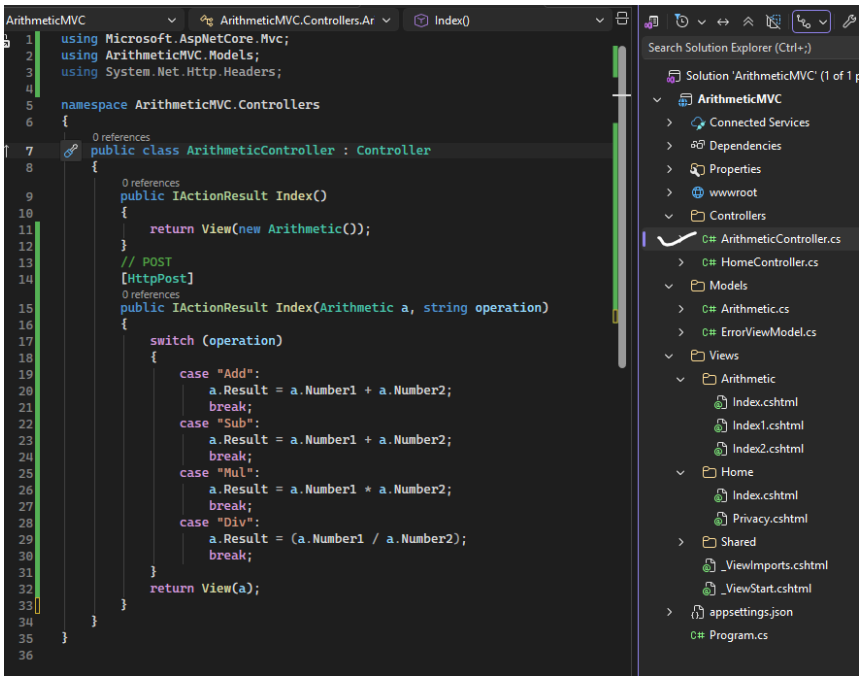
Step 5: Create the Model

1. Right-click **Models**→elect **Add** → **Class**. → Name the class **Arithmetic.cs**→ Click **Add**.



Step 6: Create the Controller

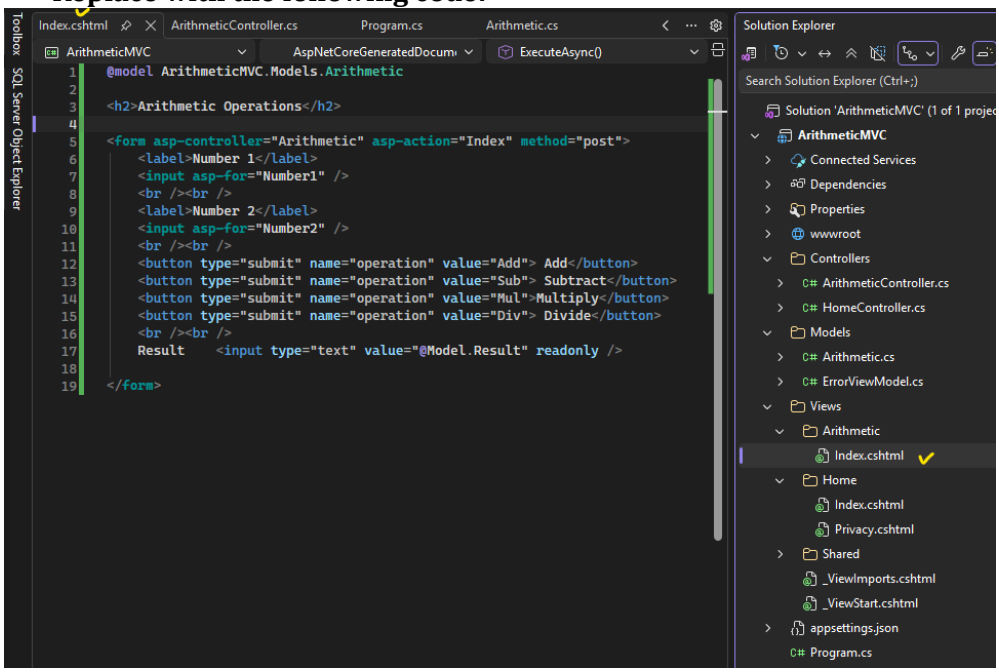
1. Right-click **Controllers** →Select **Add** → **Controller** →Choose **MVC Controller – Empty** →Click **Add**. → Enter the name **ArithmeticController** → Click **Add**.



```
using Microsoft.AspNetCore.Mvc;
using ArithmeticMVC.Models;
using System.Net.Http.Headers;

namespace ArithmeticMVC.Controllers
{
    0 references
    public class ArithmeticController : Controller
    {
        0 references
        public IActionResult Index()
        {
            return View(new Arithmetic());
        }
        // POST
        [HttpPost]
        0 references
        public IActionResult Index(Arithmetic a, string operation)
        {
            switch (operation)
            {
                case "Add":
                    a.Result = a.Number1 + a.Number2;
                    break;
                case "Sub":
                    a.Result = a.Number1 + a.Number2;
                    break;
                case "Mul":
                    a.Result = a.Number1 * a.Number2;
                    break;
                case "Div":
                    a.Result = (a.Number1 / a.Number2);
                    break;
            }
            return View(a);
        }
    }
}
```

Step 7: Right click Arithmetic folder inside View Folder->Select **Add View**. View Name: **Index**
Replace with the following code:



```
@model ArithmeticMVC.Models.Arithmetic

<h2>Arithmetic Operations</h2>

<form asp-controller="Arithmetic" asp-action="Index" method="post">
    <label>Number 1</label>
    <input asp-for="Number1" />
    <br /><br />
    <label>Number 2</label>
    <input asp-for="Number2" />
    <br /><br />
    <button type="submit" name="operation" value="Add"> Add</button>
    <button type="submit" name="operation" value="Sub"> Subtract</button>
    <button type="submit" name="operation" value="Mul">Multiply</button>
    <button type="submit" name="operation" value="Div"> Divide</button>
    <br /><br />
    Result    <input type="text" value="@Model.Result" readonly />

</form>
```

Program.cs

```
var builder = WebApplication.CreateBuilder(args);
builder.Services.AddControllersWithViews();

var app = builder.Build();
app.UseHttpsRedirection();
app.UseStaticFiles();
app.UseRouting();
app.UseAuthorization();
app.MapControllerRoute(
   name: "default",
   pattern: "{controller=Arithmetic}/{action=Index}/{id?}");

app.Run();
```